1.70 Forces: Pulleys

Worked Solutions

1 a)

On A $R(\downarrow)$: $9 \times 9.8 - T = 9a$

On B $R(\uparrow)$: $T - 3 \times 9.8 = 3a$

$T = 44.1$ N, $a = 4.9$ ms^{-2}

$T = 44$ N (2 SF – AQA), $a = 4.9$ ms^{-2} (2 SF – AQA)

1 b)

On A $R(\uparrow)$: $T - 2 \times 9.8 = 2a$

On B $R(\downarrow)$: $8 \times 9.8 - T = 8a$

$T = 31.36$ N, $a = 5.88$ ms^{-2}

$T = 32$ N (2 SF – AQA), $a = 5.9$ ms^{-2} (2 SF – AQA)

1 c)

On A $R(\uparrow)$: $T - 0.2 \times 9.8 = 0.2a$

On B $R(\downarrow)$: $0.5 \times 9.8 - T = 0.5a$

$T = 2.8$ N, $a = 4.2$ ms^{-2}

2 a)

On A $R(\downarrow)$: $5m \times 9.8 - T = 5ma$

On B $R(\uparrow)$: $T - 3m \times 9.8 = 3ma$

Adding the equations:

$49m - T + T - 29.4m = 8ma$

$49 - 29.4 = 8a \Rightarrow a = 2.45$ ms^{-2}

$a = 2.5$ ms^{-2} (2 SF – AQA)

2 b)

$T = 36.75m$ N

2 c)

$2T = 73.5m$ N

3)

On A $R(\uparrow)$: $T - mg = ma$

On B $R(\downarrow)$: $Mg - T = Ma$

Adding the equations:

$Mg - T + T - mg = ma + Ma$

$(M - m)g = (M + m)a$

$\Rightarrow a = \left(\frac{M-m}{M+m}\right) g$ as req.

4)

On A $R(\downarrow)$: $kmg - T = kma$

On B $R(\uparrow)$: $T - mg = ma$

Adding the equations:

$kmg - T + T - mg = kma + ma$

$kg - g = ka + a$

$g(k-1) = a(k+1)$

$a = \left(\frac{k-1}{k+1}\right) g$ as req.

5 a)

$R(\downarrow)$: $0.8 = 0 + \frac{1}{2}a \times 4^2 \Rightarrow a = 0.1$ ms^{-2}

5 b)

On A $R(\downarrow)$: $2 \times 9.81 - T = 2 \times 0.1$

$\Rightarrow T = 19.42$ N

$T = 19.4$ N (3 SF – AQA)

5 c)

On B $R(\uparrow)$: $19.42 - m \times 9.81 = m \times 0.1$

$\Rightarrow m = 1.96$ kg (3 SF)

6 a)

On A $R(\downarrow)$: $4 \times 9.81 - T = 4a$

On B $R(\uparrow)$: $T - 3 \times 9.81 = 3a$

$a = \frac{981}{700} = 1.40$ ms^{-2}

6 b)

$T = \frac{5886}{175} = 33.6$ N

6 c)

$R(\downarrow)$: $1.5 = 0 + \frac{1}{2} \times \frac{981}{700} \times t^2 \Rightarrow t = 1.46$ s

6 d)

$v^2 = 0 + 2 \times \frac{981}{700} \times 1.5$

$\Rightarrow v = \sqrt{\frac{2943}{700}} = 2.050435494\ldots$

When A hits the ground, B continues upwards until it slows to 0 ms^{-1}

$v^2 = u^2 + 2as$

$0 = \frac{2943}{700} + 2 \times -9.81s \Rightarrow s = \frac{3}{14} = 0.2142857\ldots$

Therefore 1.71 metres above the floor.

Modelling assumptions: there are no resistances to motion (such as friction or air resistance), the pulley is small, particle B does not hit the pulley.

7 a)

$R(\uparrow)$: $0.2 = 0 + \frac{1}{2}a \times 0.5^2 \Rightarrow a = 1.6$ ms^{-2}

7 b)

On A $R(\uparrow)$: $T - 5 \times 9.8 = 5 \times 1.6 \Rightarrow T = 57$ N

7 c)

On B $R(\downarrow)$: $m \times 9.8 - 57 = m \times 1.6$

$\Rightarrow m = \frac{285}{41} = 6.951219512 \approx 7$ kg as required.

7 d)

$v^2 = 0 + 2 \times 1.6 \times 0.5 \Rightarrow v = \frac{2}{5}\sqrt{10} = 1.26$ ms^{-1}

7 e)

$0 = 1.6 + 2 \times -9.8 \times s$

$\Rightarrow s = \frac{4}{49} = 0.08163265306\ldots$ m

$1 + 0.5 + 0.08163\ldots = 1.58163\ldots$

Therefore 1.58 metres above the floor.

7 f)

$0 = \frac{2}{5}\sqrt{10}t + \frac{1}{2} \times -9.8 \times t^2$

$\Rightarrow t = 0$ and $t = 0.2581451151$

So 0.258 seconds

0.26 seconds (2 SF – AQA)

7 g)

Box A will ascend with a higher acceleration and there will be a higher tension in the string.

7 h)

Box A will not move as $a = 0$ and so the tension will be equal to the weight of box A

7 i)

Box A will descend and there will be a lower tension in the string.

8 a)

On A $R(\downarrow)$: $3 \times 9.8 - T = 3a$

On B $R(\uparrow)$: $T - 0.5 \times 9.8 = 0.5a$

$a = 7$ ms^{-2}

8 b)

$T = 8.4$ N

8 c)

$s = 0 + \frac{1}{2} \times 7 \times t^2$

$s = 3.5t^2$

Distance of $2h$ metres

$\Rightarrow 2h = 3.5t^2$

$\frac{4}{7}h = t^2 \Rightarrow t = \sqrt{\frac{4h}{7}}$ seconds as req.

(Time must be positive)

9 a)

On B $R(\uparrow)$: $T - 8 \times 9.8 = 8 \times 1.96 \Rightarrow T = 94.08$ N

On B $R(\uparrow)$: $T - 8 \times 9.8 = 0 \Rightarrow T = 78.4$ N

So $78.4 < T \leq 94.08$

$78 < T \leq 94$ (2 SF – AQA)

9 b)

$m > 6$

On A $R(\downarrow)$: $(m + 2) \times 9.8 - 94.08 = (m + 2) \times 1.96$

$9.8m + 19.6 - 94.08 = 1.96m + 3.92$

$7.84m = 78.4 \Rightarrow m = 10$

$6 < m \leq 10$

10 a)

On A $R(\rightarrow)$: $T = 1.2a$

On B $R(\downarrow)$: $0.5 \times 9.8 - T = 0.5a$

$a = \frac{49}{17} = 2.88$ ms^{-2}

$a = 2.9$ ms^{-2} (2 SF – AQA)

10 b)

$T = 3.46$ N

$T = 3.5$ N (2 SF – AQA)

11 a)

On A $R(\rightarrow)$: $T - 20 = 2a$

On B $R(\downarrow)$: $3 \times 9.8 - T = 3a$

$a = 1.88$ ms^{-2}, $T = 23.76$ N

$a = 1.9$ ms^{-2}, $T = 24$ N (2 SF – AQA)

11 b)

On A $R(\rightarrow)$: $T - 12 = 5a$

On B $R(\downarrow)$: $7 \times 9.8 - T = 7a$

$a = 4.72$ ms^{-2}, $T = 35.6$ N

$a = 4.7$ ms^{-2}, $T = 36$ N (2 SF – AQA)

11 c)

On A $R(\rightarrow)$: $T - 2 = 0.2a$

On B $R(\downarrow)$: $0.5 \times 9.8 - T = 0.5a$

$a = 4.14$ ms^{-2}, $T = 2.83$ N

$a = 4.1$ ms^{-2}, $T = 2.8$ N (2 SF – AQA)

11 d)

On A $R(\rightarrow)$: $T - 58.8 = 8a$

On B $R(\downarrow)$: $6 \times 9.8 - T = 6a$

$a = 0$ ms^{-2}, $T = 58.8$ N

$a = 0$ ms^{-2}, $T = 59$ N (2 SF – AQA)

12 a)

$1 = 0 + \frac{1}{2} \times \frac{47}{25} \times t^2 \Rightarrow t = 1.03$ s

$t = 1.0$ s (2 SF – AQA)

12 b)

$1 = 0 + \frac{1}{2} \times \frac{283}{60} \times t^2 \Rightarrow t = 0.651$ s

$t = 0.65$ s (2 SF – AQA)

12 c)

$1 = 0 + \frac{1}{2} \times \frac{29}{7} \times t^2 \Rightarrow t = 0.695$ s

$t = 0.69$ s (2 SF – AQA)

12 d)

The system does not move

13 a)

On A $R(\leftarrow)$: $2500 - 450 - T = 2000a$

On B $R(\uparrow)$: $T - 100 \times 10 = 100a$

$a = 0.5$ ms^{-2}

13 b)

$T = 1050$ N

$T = 1000$ N (1 SF – AQA)

13 c)

The cable is light and inextensible. The pulley is small.

13 d)

$s = 0 + \frac{1}{2} \times 0.5 \times 10^2 = 25$ metres

13 e)

No resistances acting against the pallet of building materials.

14)

On A $R(\rightarrow)$: $T = Ma$

On B $R(\downarrow)$: $10m - T = ma$

$T + 10m - T = Ma + ma$

$10m = (m + M)a$

$a = \frac{10m}{m+M}$ as req.

15 a)

On A $R(\rightarrow)$: $T - 5 = 0.3a$

On B $R(\downarrow)$: $9.8m - T = ma$

$-5 + 9.8m = 0.3a + ma$

$-5 + 9.8m = a(0.3 + m)$

$a = \frac{9.8m-5}{m+0.3}$

15 b)

$1.2 = 0 + \frac{1}{2}a \times 4 \Rightarrow a = 0.6$ ms^{-2}

$-5 + 9.8m = 0.3 \times 0.6 + 0.6m$

$9.2m = 5.18 \Rightarrow m = \frac{259}{460} = 0.563$ kg

$m = 0.56$ kg (2 SF – AQA)

15 c)

There are no resistances to motion on block B

16 a)

On A: $R(\leftarrow)$: $T_1 - T_2 = 2a$

On B: $R(\downarrow)$: $8 \times 9.8 - T_1 = 8a$

On C: $R(\uparrow)$: $T_2 - 3 \times 9.8 = 3a$

$a = \frac{49}{13} = 3.77$ ms^{-2}

$a = 3.8$ ms^{-2} (2 SF – AQA)

16 b)

$T_1 = \frac{3136}{65} = 48.2$ N, $T_2 = \frac{2646}{65} = 40.7$ N,

$T_1 = 48$ N (2 SF – AQA), $T_2 = 41$ N (2 SF – AQA)

17)

On A $R(\rightarrow)$: $T - \frac{1}{2}kmg = ma$

On B $R(\downarrow)$: $kmg - T = kma$

$T + kmg - T - \frac{1}{2}kmg = ma + kma$

$\frac{1}{2}kmg = (1+k)ma$

$\frac{1}{2}kg = (1+k)a$

$a = \frac{kg}{2(1+k)}$ as req.

18 a)

$0.5 = 0 + \frac{1}{2}a \times 100 \Rightarrow a = 0.01$ ms^{-2}

18 b)

On A $R(\leftarrow)$: $20mg - T - 10mg = 0.03m$

On B $R(\uparrow)$: $T - kmg = 0.01km$

$20mg - T - 10mg + T - kmg = 0.03m + 0.01km$

$20g - 10g - kg = 0.03 + 0.01k$

$10g - 0.03 = kg + 0.01k$

$10g - 0.03 = k(g + 0.01)$

$k = \frac{10g-0.03}{g+0.01} = \frac{1000g-3}{100g+1}$